



Assignment 1

Objective:

- Control a DC motor to rotate clockwise for 5 seconds, then counter-clockwise for 3 seconds, then stop for 5 seconds, repeating the cycle using a timer.

Components:

- ATmega32 microcontroller
- DC motor
- Resistors
- wires

Circuit Setup:

- Connect the DC motor to the motor driver, which is controlled by the ATmega32.

Programming:

- Initialize the timer to generate interrupts or delays for accurate timekeeping.
- Rotate the motor clockwise for 5 seconds.
- Reverse the motor to rotate counter-clockwise for 3 seconds.
- Stop the motor for 5 seconds.